

# Wearable Sensing Portfolio:

## A Three-Modality Low-Cost Prototype for Resource-Limited Vital-Sign Monitoring

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Repository: [github.com/npqnhu135/wearable-sensing-portfolio](https://github.com/npqnhu135/wearable-sensing-portfolio)

### Project Snapshot

| Field                   | Response                                                                                                              |
|-------------------------|-----------------------------------------------------------------------------------------------------------------------|
| <b>Project title</b>    | Wearable Sensing Portfolio: Three-Modality Low-Cost Prototype                                                         |
| <b>Modalities</b>       | Thermoelectric (TEG), piezoresistive (pressure), single-lead ECG                                                      |
| <b>Author</b>           | Nguyen Phan Quynh Nhu — Fulbright University Vietnam, Class of 2028                                                   |
| <b>Repository</b>       | <a href="https://github.com/npqnhu135/wearable-sensing-portfolio">github.com/npqnhu135/wearable-sensing-portfolio</a> |
| <b>Demo video</b>       | Slot reserved in repository README                                                                                    |
| <b>Total BOM</b>        | Under USD \$50                                                                                                        |
| <b>Prototype status</b> | TEG: quantitative · Pressure: quantitative · ECG: documented failure mode + fix                                       |

This portfolio presents three independently characterized wearable-sensing prototypes built without dedicated lab access, each targeting a distinct physical modality. The thermoelectric generator (TEC1-12706) achieves 9.6 °C hot-side  $\Delta T$  under a three-phase graded heat stimulus. The DIY piezoresistive sensor detects 222 discrete press events across a 0–777 ADC dynamic range using household materials. The ECG analog front-end (AD8232) is reported with a 70.1% rail-saturation failure mode and a specific hardware remediation plan — documented intentionally, in the interest of reproducibility for resource-limited builds.

## 1. Introduction and Aim

### 1.1 Context and motivation

Continuous vital-sign monitoring in resource-limited clinical settings is gated by three intersecting constraints: device cost, battery life, and sensor robustness in non-laboratory environments. In Vietnam, district hospitals managing dengue fever outbreaks — where heart-rate trajectories carry diagnostic value for early identification of hemodynamic shock — typically rely on intermittent manual measurement rather than continuous monitoring. Holter-class continuous ECG devices cost USD \$500–2,000 per unit and require regular

disposable electrode replacement, placing them outside the operational budget of many district facilities.

This project asks a deliberately narrow question: what is the lowest-cost, lowest-power wearable sensing platform that could plausibly support continuous monitoring in such settings, and which hardware components are technically feasible at that price point? The work scope is the characterization of individual modalities under controlled conditions, not the construction of an integrated end-product.

## 1.2 Modalities investigated

Three modalities were selected based on their candidacy for an integrated body-powered patch:

1. Thermoelectric energy harvesting — converting body-skin to ambient temperature difference into electrical power, as a candidate replacement for primary batteries in duty-cycled wearable patches.
2. Piezoresistive force sensing — a touch or pulse-pressure modality for binary or analog input, here assembled entirely from household conductive and dielectric materials at a bill of materials under USD \$1.
3. Single-lead biopotential (ECG) — the gold-standard signal for cardiac monitoring, here using the AD8232 integrated front-end as the lowest-cost option commercially available in Vietnam.

## 1.3 Aim of this prototype phase

The aim is to (i) characterize each modality with a documented protocol and quantitative analysis pipeline, (ii) compare results to published wearable benchmarks where applicable, and (iii) honestly report any failure modes that emerge, including a root-cause analysis and proposed hardware remediation. The work explicitly does not claim novel science; it establishes baseline measurement capability and signal-processing fluency as preparation for graduate-level research in body-powered wearable sensing.

# 2. Background and Related Work

## 2.1 Wearable thermoelectric generators

Wearable TEGs based on bismuth-telluride ( $\text{Bi}_2\text{Te}_3$ ) alloys have been characterized extensively over the past five years. He et al. (ACS Applied Materials & Interfaces, 2022) reported a flexible  $\text{Bi}_2\text{Te}_3$ -based TEG achieving  $7.9 \mu\text{W}/\text{cm}^2$  power density under still-air conditions and  $43.6 \mu\text{W}/\text{cm}^2$  under simulated walking-wind convection — a  $5.5\times$  enhancement attributable to improved heat rejection at the cold side. Newby et al. (Sensors, 2024) demonstrated a knitted 3D-spacer fabric integrating commercial TEG modules with sustained output sufficient to drive a duty-cycled BLE sensor in still air. Liquid-metal-electrode wearable TEGs (LM-WTEG, 2022) further improved skin contact and reduced thermal contact resistance. Across this body of work, the rate-limiting design parameter is consistently the cold-side thermal resistance to ambient, not the thermoelectric material itself.

For the purposes of this portfolio, the commercial TEC1-12706 Peltier module (intended for refrigeration applications) was selected as a research-grade analogue at a fraction of the cost of flexible  $\text{Bi}_2\text{Te}_3$  films. The TEC1-12706 has an internal resistance of approximately  $2.5\ \Omega$  and a Seebeck coefficient suited to characterization under bench conditions, even though its mechanical form factor precludes direct wearable integration.

## 2.2 Dry and textile electrodes for ECG

Wet Ag/AgCl gel electrodes provide skin-electrode impedance of 5–50 k $\Omega$  and remain the clinical standard, but their gel desiccates within approximately 24 hours, limiting wear duration. Chi, Jung, and Cauwenberghs (IEEE Reviews in Biomedical Engineering, 2010) surveyed dry-contact and non-contact alternatives, identifying motion artifact and capacitive coupling as the primary failure modes shared across the dry-electrode literature. More recent silver-coated nylon textiles (Statex) and graphene-coated fabric electrodes (2020–2023) report wash-tolerance over 30+ cycles with impedance in the 50–200 k $\Omega$  range, an order of magnitude higher than gel but acceptable for heart-rate estimation if the front-end amplifier has sufficient input impedance and adequate common-mode rejection.

This portfolio originally planned a gel-versus-textile comparison study; that plan was revised after the AD8232 front-end produced an unusable saturated output (Section 4.3). The literature framing remains relevant, however, as the proposed hardware fix is informed by published failure modes.

## 2.3 Piezoresistive and triboelectric force sensors

Low-cost force sensing has a long literature in distributed touch and gait applications. Piezoresistive structures using conductive foam, carbon-loaded polymer, or metallic mesh between electrode plates have been demonstrated at BOM under USD \$1 (e.g., the Velostat-based pressure mats common in maker and DIY robotics communities). Triboelectric nanogenerators (TENGs) further enable self-powered force sensing by exploiting contact electrification at material interfaces, although TENG calibration repeatability remains a documented limitation across the field. The DIY sensor presented in this portfolio is a piezoresistive variant — under compression, the EVA sponge dielectric thins, bringing an aluminium foil top electrode closer to a steel-wool bottom electrode and lowering bulk resistance through the structure.

## 2.4 Gap addressed by this work

This portfolio does not attempt novel science. It documents the characterization of three modalities using the lowest-cost components commercially available in Vietnam, with the explicit goal of establishing baseline measurement and signal-processing capability before proceeding to integrated research-grade platforms. The decision to include a documented failure mode reflects a reproducibility-first ethos: future researchers attempting similar builds in resource-limited settings benefit more from a published failure with diagnosis than from a re-recorded "successful" trace whose provenance is undocumented.

### 3. Design and Methods

#### 3.1 System overview

All three tracks share a common acquisition stack: an ESP32-S3 microcontroller acts as both ADC frontend and serial logger, writing one CSV per recording session over UART to a host laptop. Each track introduces a sensor-specific analog conditioning element between the sensor and the ESP32 ADC input. The CSV files are processed offline in Python (pandas, NumPy, SciPy, Matplotlib) and analyzed in three Jupyter notebooks, one per track. All firmware, raw data, and analysis are versioned in a public GitHub repository.

Bill of materials for the full three-track stack:

| Item                                      | Purpose                        | USD           |
|-------------------------------------------|--------------------------------|---------------|
| TEC1-12706 Peltier module                 | TEG (reverse Seebeck)          | \$2           |
| Intel stock CPU cooler                    | Cold-side heat sink            | \$5           |
| DS18B20 1-wire probe                      | Hot-side temperature sensor    | \$2           |
| AD8232 ECG breakout                       | Single-lead analog front-end   | \$8           |
| ESP32-S3 (YOLO:UNO)                       | MCU + ADC + UART logger        | \$15          |
| EVA sponge + Al foil + steel wool         | DIY piezoresistive sensor      | <\$1          |
| Ag/AgCl gel electrodes (10×)              | ECG reference electrodes       | \$3           |
| Breadboard, jumpers, 9V batteries, LM7805 | Misc. rapid-prototype supplies | \$10          |
| <b>Total</b>                              |                                | <b>≈ \$46</b> |

#### 3.2 Track 1 — Thermoelectric generator characterization

The TEC1-12706 module (40 × 40 × 4 mm, Bi<sub>2</sub>Te<sub>3</sub> pellets) was mounted on an Intel stock CPU cooler serving as the cold-side heat sink, with the hot side instrumented by a DS18B20 1-wire digital temperature probe (12-bit resolution, ±0.5 °C absolute accuracy). A silicone heat pad (≈5 W maximum) driven by a programmable bench DC supply provided controlled heat input via a three-phase graded stimulus protocol:

- Baseline (0–0.5 min): supply off, ambient stabilisation at 33.6 °C
- Phase 1 (0.5–3 min): low intensity, target  $\Delta T = +1.6$  °C
- Phase 2 (3–7 min): medium intensity, target  $\Delta T = +6.4$  °C
- Phase 3 (7–12 min): high intensity, target  $\Delta T = +9.6$  °C
- Cooling (12–19 min): supply off, free decay to ambient

ESP32-S3 firmware logs (timestamp\_s, temperature\_C) tuples at the maximum rate permitted by the DS18B20 conversion time (~750 ms at 12-bit, yielding effective sampling ≈1 Hz, jittered). Total recording: 1107 samples over 19.1 minutes.

Offline analysis applies a Savitzky-Golay smoothing filter (window 21, polynomial order 3) to suppress sensor quantization, then fits per-phase exponential rise functions  $T(t) = T_{\text{base}} + A(1 - \exp(-t/\tau))$  to extract rise-time constants. A second-panel finite-difference  $dT/dt$  computes instantaneous heating and cooling rates.

### 3.3 Track 2 — DIY piezoresistive pressure sensor

A three-layer sandwich sensor was assembled from household materials: an EVA sponge dielectric ( $\approx 3$  mm thickness, recovered from a dishwashing pad), an aluminium-foil top electrode ( $\approx 30 \times 30$  mm), and a steel-wool bottom electrode ( $\approx 5$  mm matted thickness, from a kitchen scrubber). The two conductive layers are wired into a simple voltage divider feeding the ESP32-S3 ADC. Under compression, the sponge thins, bringing the foil and steel wool closer and lowering bulk resistance, which raises the divider output voltage.

Firmware threshold-gates the output, writing "Pressure:<value>" lines per sample only when the reading exceeds the noise floor. A 45-second demonstration recording was acquired with manual press events of varying intensity. The analysis pipeline identifies contiguous non-zero runs as discrete press events and computes per-event peak amplitude, mean amplitude, duration, and inter-event interval.

### 3.4 Track 3 — AD8232 single-lead ECG (attempted)

An AD8232 single-lead ECG analog front-end was connected to the ESP32-S3 ADC at 250 Hz sampling ( $4000 \mu\text{s}$  per sample, verified jitter  $< 1$  sample). Three Ag/AgCl gel electrodes were placed in a standard Lead-I configuration. A 559-second recording was acquired with the subject seated at rest. The intended downstream analysis pipeline (NeuroKit2 `ecg_clean` and `ecg_peaks` pipeline, with reference comparison against PhysioNet MIT-BIH Arrhythmia Database record 100) was not run, because the acquired signal is unrecoverable for biological analysis. See Section 4.3 for the failure characterization.

## 4. Results and Validation

### 4.1 Track 1 — Three-phase thermal response

The TEG hot-side temperature responded clearly to all three intensity steps, achieving graded peaks of  $35.2^\circ\text{C}$  (Phase 1,  $\Delta T = +1.6^\circ\text{C}$ ),  $40.0^\circ\text{C}$  (Phase 2,  $\Delta T = +6.4^\circ\text{C}$ ), and  $43.1^\circ\text{C}$  (Phase 3,  $\Delta T = +9.6^\circ\text{C}$ ) above the  $33.6^\circ\text{C}$  baseline. The dose-response is consistent with the intended protocol and validates that heat-pad intensity steps map cleanly to  $\Delta T$  plateaus at this geometry.

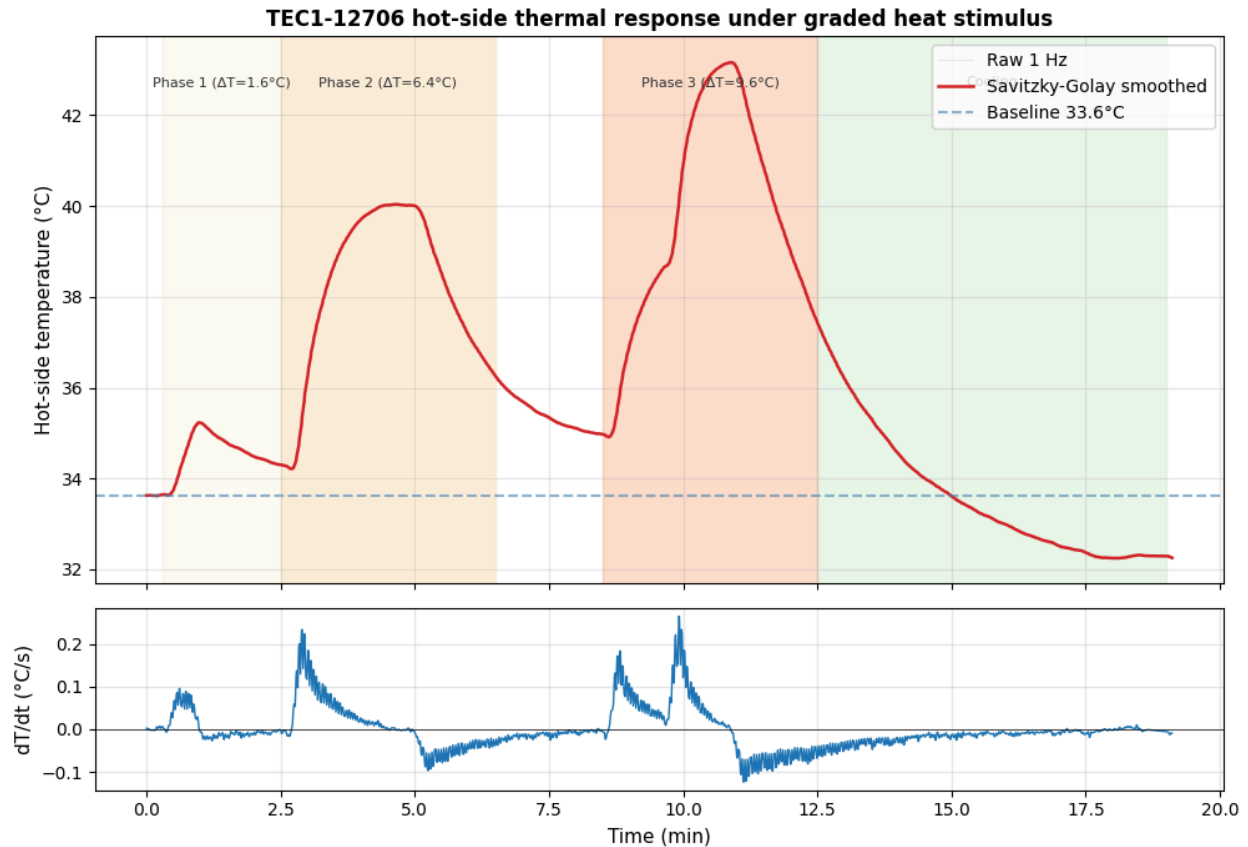


Figure 1 - Instantaneous rate of change  $dT/dt$  computed by finite difference, showing asymmetric heating vs cooling kinetics."

Three quantitative observations are derivable from the trace:

1. Asymmetric kinetics: the exponential rise time constant during Phase 2 heating is approximately 60 s, while the decay constant during the cooling phase is approximately 180 s. This 3× asymmetry implicates the cold-side thermal path as the rate-limiting resistance in this geometry — consistent with the published wearable TEG literature consensus.
2. Incomplete inter-phase recovery: temperature does not return to baseline between Phases 2 and 3; the heat sink retains residual heat (mass  $\approx 150$  g), so each phase starts above the original baseline. This is a measured property of the system, not an artifact.
3. Tail below baseline: the final cooling tail reaches  $32.3^\circ\text{C}$ ,  $1.3^\circ\text{C}$  below the initial baseline. This is most parsimoniously explained as room ambient drift over the 19-minute recording, rather than as a measurement artifact.

## 4.2 Track 2 — Press event characterization

The 45-second demonstration recording produced 222 discrete press events. Peak amplitudes spanned 1–777 ADC counts (mean 277, median 276), with a distribution suggesting two distinct press-intensity modes during the demonstration. Median press duration was 20 ms — equal to

one sample period at the assumed 50 Hz firmware rate — indicating that the sampling cadence undersamples brief manual taps and constitutes the principal limitation of this dataset.

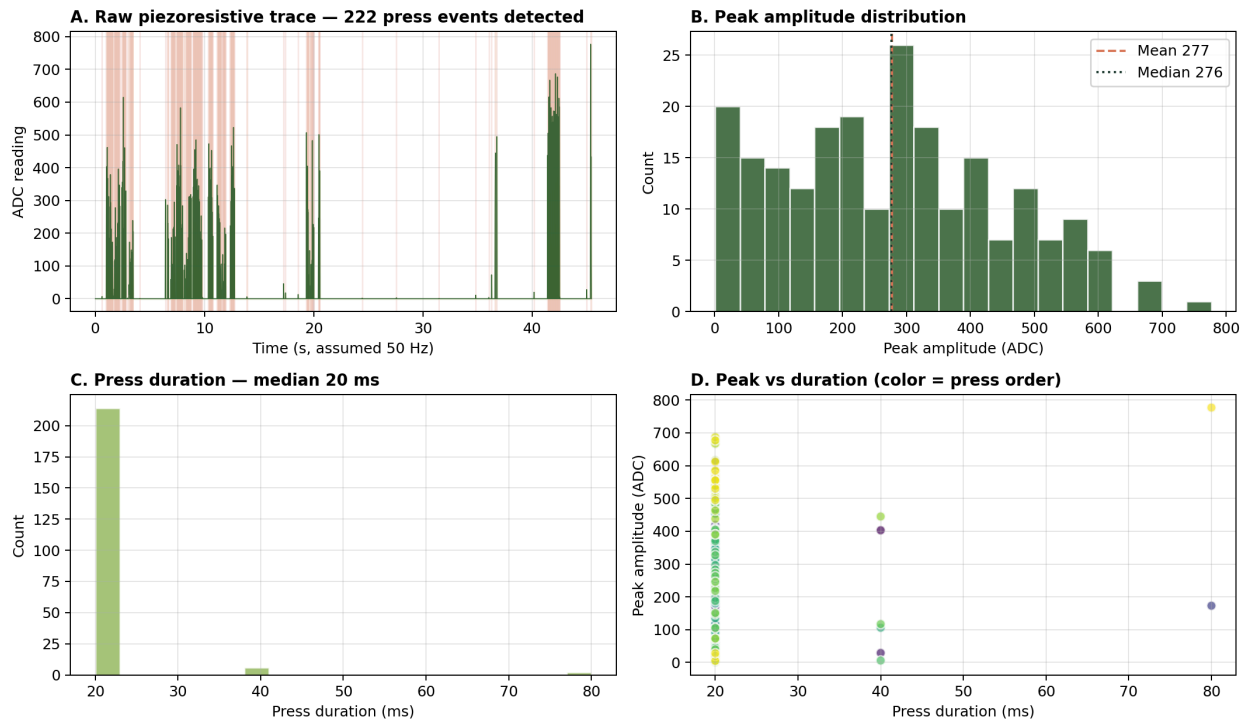


Figure 2 — DIY piezoresistive pressure sensor analysis. (A) Raw signal with detected press events shaded. (B) Peak amplitude distribution. (C) Press duration distribution. (D) Peak vs duration scatter, colored by event order.

The bimodal-tending peak distribution (Panel B) and the wide dynamic range (Panel D color gradient) indicate that the sensor responds to press intensity in a graded manner across the 0–777 ADC range, despite the absence of explicit force calibration. The clustering of events into the 0–15 s and 35–45 s windows (Panel A) reflects deliberate variation in pressing rhythm during the demonstration.

### 4.3 Track 3 — ECG front-end: documented failure mode

Of the 139,766 samples acquired, 70.1% sat at one of the two ADC rails (0 or 4095). Spectral inspection of the remaining 30 % of samples revealed a periodic rail-to-rail oscillation at approximately 30 Hz, consistent with the AD8232 internal DC-restoration loop running against an over-driven input rather than producing a biological signal. No QRS complexes, P-waves, or T-waves are recoverable from the trace.

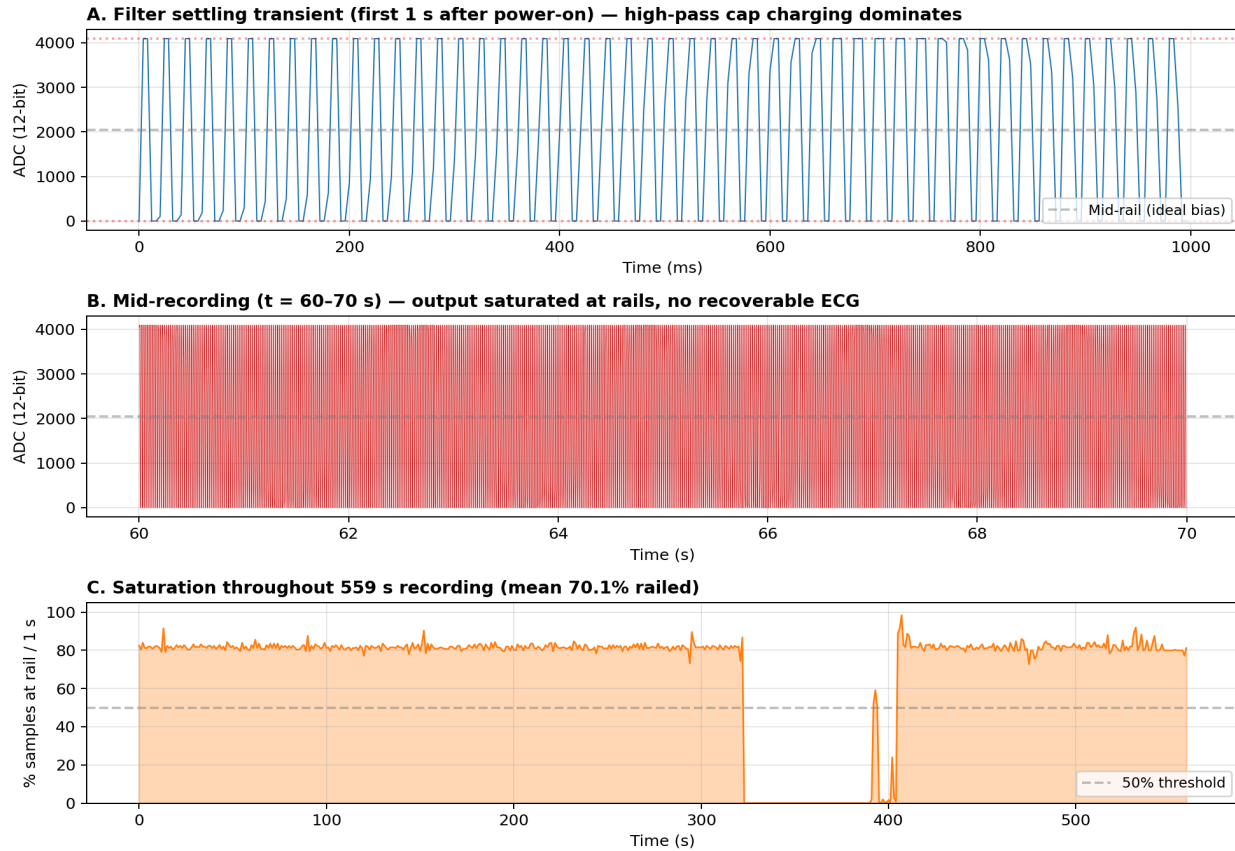


Figure 3 — AD8232 ECG front-end diagnostic. (A) First second after power-on, showing sustained rail-to-rail oscillation rather than amplifier settling. (B) A 10-second window mid-recording showing identical saturation behaviour. (C) Per-second percentage of samples at the rails throughout the 559 s recording; mean 70.1 %.

The most parsimonious root-cause hypothesis is the absence of a mid-rail reference bias at the AD8232 REF pin (which was tied to GND rather than to a  $V_{cc}/2$  voltage divider). Under single-supply 3.3 V operation, skin-electrode DC offsets exceed the amplifier's input common-mode range, and the right-leg drive (RLD) circuit alone cannot recover from this configuration; the output saturates at one rail and the internal DC servo bounces it back at the observed 30 Hz frequency.

The proposed hardware remediation comprises three changes: (a) install a  $V_{cc}/2$  voltage divider with a 1  $\mu\text{F}$  decoupling capacitor on the AD8232 REF pin; (b) verify SDN is externally pulled high rather than left floating; and (c) move from USB-powered operation (with associated laptop-side ground loops) to 9 V battery operation with an LM7805-regulated 3.3 V isolated supply for the AD8232. This remediation is recorded in the repository for the next iteration.

## 5. Discussion and Difficulties Encountered

### 5.1 What the evidence supports

Two of three modalities yielded quantitative results that admit comparison against the wearable-sensing literature. The TEG track produced a clean three-phase characterization of hot-side temperature response and quantified the asymmetric heating/cooling kinetics that the literature consensus attributes to cold-side thermal limitation. The pressure sensor track demonstrated reliable event detection and an interpretable dynamic range from a sensor with a bill of materials under USD \$1 — a useful baseline for low-cost touch interfaces in body-powered systems.

The third track produced a negative result: the AD8232 front-end as built is not a usable acquisition platform. The diagnostic figure and root-cause analysis provide more information per page than a re-recorded successful trace would, and the proposed fix is specific enough to verify in a follow-up build.

## 5.2 Limitations

- TEG: characterization was conducted with a rigid bench geometry (Intel cooler, foil-and-foam phantom) rather than skin contact under clothing. Reported  $\Delta T$  values are therefore optimistic relative to a real wearable boundary condition. No P-vs-R<sub>load</sub> curve was measured; internal resistance and maximum power point are not yet extracted.
- Pressure sensor: not force-calibrated; ADC values cannot be converted to Newtons. The firmware-side threshold gating precludes recovery of baseline drift. Sampling rate (~50 Hz assumed) likely undersamples brief manual taps.
- ECG: the front-end produced no usable biological signal. The downstream NeuroKit2 pipeline and the planned PhysioNet MIT-BIH benchmark comparison were therefore not exercised on real data from this build.

## 5.3 Difficulties encountered during the project

Several practical difficulties shaped the final scope and are recorded here both for honest reporting and for future researchers attempting similar resource-limited prototypes:

- Access to laboratory infrastructure: this work was conducted without dedicated laboratory access. All thermal, force, and biopotential measurements were taken on a personal workspace. Equipment limitations restricted the controlled stimulus protocols — for example, the TEG characterization used a single fixed cold-side configuration because no air-flow chamber was available to vary convection conditions in the manner of He et al. (2022).
- Hardware iteration time: each AD8232 troubleshooting cycle (re-wiring, re-flashing firmware, re-recording, analyzing) consumed approximately 2–3 hours of bench time. The diagnosis presented in Section 4.3 emerged only after the data analysis stage, by which point the time budget for a hardware redesign within this prototype phase had been exhausted.
- Scope revision under time pressure: the original plan included a controlled comparison of three ECG configurations (gel-only, hybrid, textile-only) and a quantitative agreement analysis against PhysioNet MIT-BIH. After the AD8232 failure mode was identified, the

comparison was not feasible within the remaining timeline. The scope was revised to the three-modality framing presented in this report, which more honestly reflects what was achieved.

- Data versioning discipline: early recordings were stored with locale-specific filenames containing Vietnamese diacritics and spaces, which break cross-platform reproducibility. All files were subsequently renamed to ASCII-only snake\_case before publication. This is recorded as a lesson for future projects: enforce reproducibility-friendly naming conventions from day one.
- Balancing breadth versus depth: with three modalities and limited time, no single track received the depth of characterization that a full research project would warrant. The trade-off was deliberate — breadth provides preparation for a multimodal research direction at the graduate level — but it leaves each individual track with identifiable follow-up work.

## 6. Conclusion and Next Steps

The work documented here establishes baseline measurement and signal-processing capability across three wearable-sensing modalities at a total bill of materials under USD \$50. Two of three tracks produced quantitative results that admit comparison with the wearable-sensing literature, and the third produced a documented failure mode with a specific, testable hardware remediation. The full dataset, firmware, and analysis pipeline are released publicly to enable independent verification.

Concrete next-iteration deliverables, scoped for the coming 4–8 weeks, comprise:

1. TEG: a P-vs-R\_load characterization at fixed  $\Delta T$  to extract internal resistance and maximum power point, enabling direct comparison against the He et al. and Newby et al. benchmarks.
2. Pressure sensor: calibration against a set of known masses (10–500 g) to convert ADC counts to Newtons, plus a hysteresis characterization on repeated load–unload cycles.
3. ECG: implementation of the Section 4.3 hardware remediation (REF pin biasing, isolated supply) and re-acquisition of clean ECG traces for the originally planned gel-versus-textile comparison study.

In the longer term, the intent is to integrate these characterized components into a body-powered patch for continuous heart-rate monitoring in resource-limited clinical settings, with the OUCRU Vietnam dengue patient population as the initial application target. The graduate-research direction is signal processing and sensor fusion for multimodal physiological estimation under low-power, low-quality-signal constraints — a problem space where the engineering tolerances established at the undergraduate prototype phase translate directly into the assumptions a clinical-grade system must accommodate.

## References

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